



# “ WE DECLARE THAT THE PROPER USE OF SCIENCE IS NOT TO CONQUER NATURE BUT TO LIVE IN IT. ”

Professor Barry Commoner, American biologist, Earth Day, 1970

Earth Day protests like this one in New York appealed for changes in government policy that would promote a sustainable way of life.

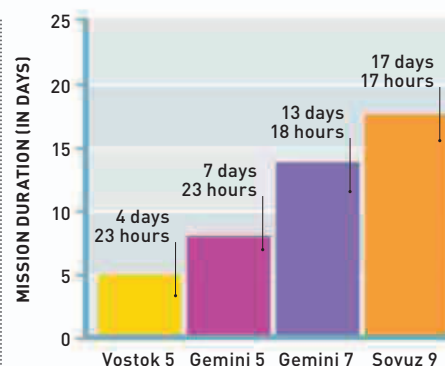
geneticist Howard Temin (1934–94) and American biologist David Baltimore (b.1938) independently discovered that **some viruses carry information in RNA**, which is then passed to **DNA**. These retroviruses include the human immunodeficiency

virus (HIV), which causes AIDS (see 1982). The enzyme involved in the process is called **reverse transcriptase**. For their work, Temin and Baltimore jointly won the 1975 Nobel Prize in Physiology or Medicine.

In July, **Type II restriction enzymes** that cut DNA into fragments at specific

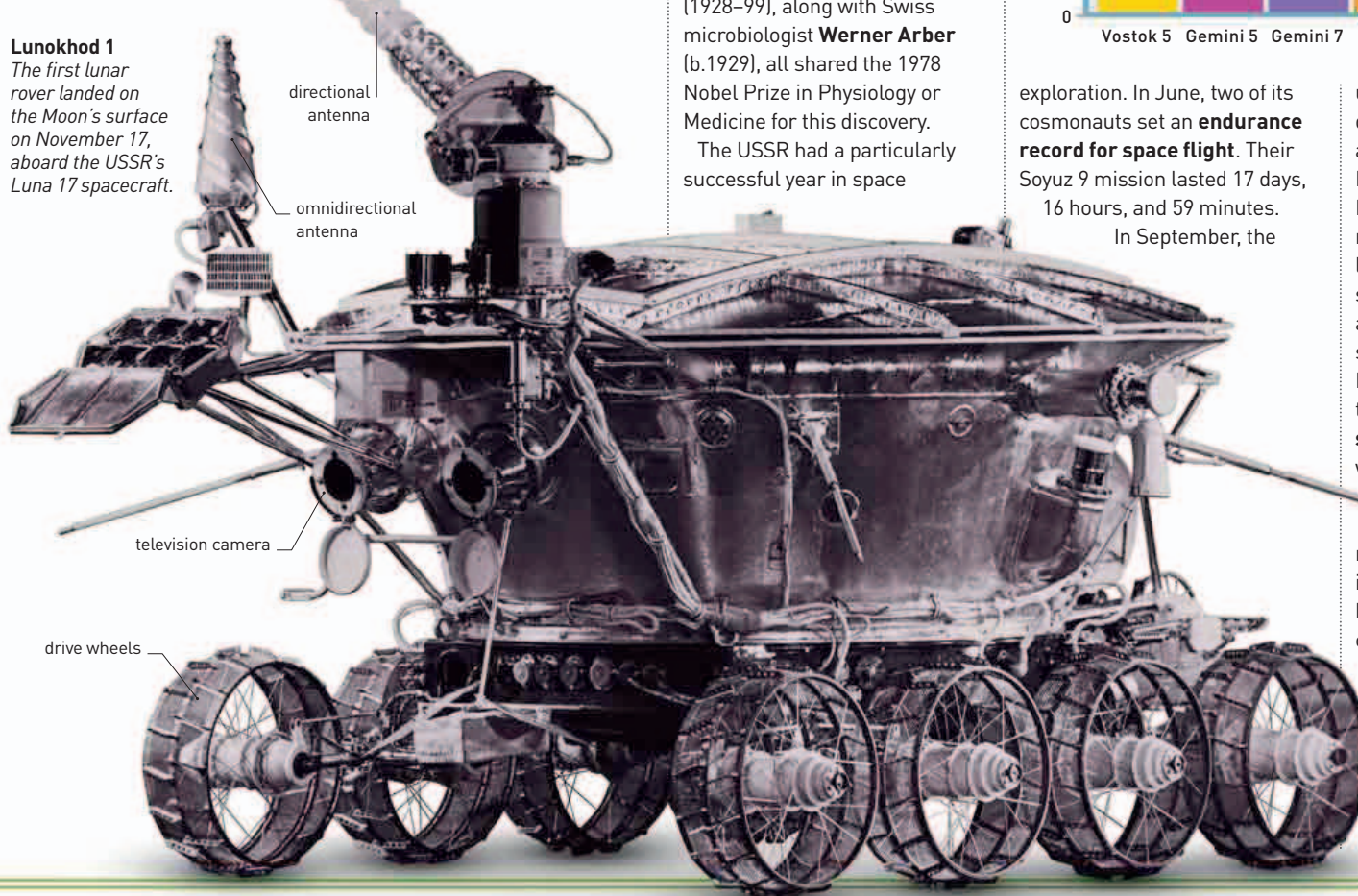
points were discovered. Restriction enzymes play a central role in modern genetic technologies, including DNA profiling (see 1984), which is used in fields such as paternity testing, criminal investigation, and ecological studies. American microbiologists **Hamilton Smith** (b.1931) and **Daniel Nathans** (1928–99), along with Swiss microbiologist **Werner Arber** (b.1929), all shared the 1978 Nobel Prize in Physiology or Medicine for this discovery.

The USSR had a particularly successful year in space



**Space missions**  
Endurance records for manned space flight steadily improved between 1963 and 1970. The Soyuz 9 record was broken in 1971, when cosmonauts on board Soyuz 11 remained in orbit for 24 days.

**Lunokhod 1**  
The first lunar rover landed on the Moon's surface on November 17, aboard the USSR's Luna 17 spacecraft.



exploration. In June, two of its cosmonauts set an **endurance record for space flight**. Their Soyuz 9 mission lasted 17 days, 16 hours, and 59 minutes. In September, the

unmanned lunar probe **Luna 16** drilled into the Moon's surface and returned samples to Earth. In November, the unmanned Luna 17 mission put the first rover—the Lunokhod—on the lunar surface. The Lunokhod, spent nearly 11 months analyzing the lunar soil and sending back photographs. In December, **Venera 7** became the first spacecraft to **make a soft landing on another planet**, when it touched down on Venus.

Engineers back on Earth detected no signals from the probe after its landing and assumed it had been destroyed. Later analysis of their recordings revealed 23 minutes of data, which indicated that the surface temperature was extremely hot at 887°F (475°C).

**July 28** Isolation of the first Type II restriction enzyme

**August 7** Corning Glass scientist Don Keck makes **optical fibers** a practical reality

**September 20** Soviet Luna 16 drills into the Moon's surface

**November 17** Soviet Lunokhod rover begins 11 months of lunar exploration

**December 12** The US launches Uhuru, the first orbiting X-ray observatory

**December 15** Soviet Venera 7 lands on Venus